

Test cases for preemptive Catering example

Parameter	Case 1: Run 1	Case 2: Run 1	Case 3: Run 1	Case 4: Run 1
Student Queue Average	31395 students	100142 students	45115 students	53414 students
Student Queue Maximum	62643 students	200021 students	90409 students	107165 students
Student Queue Variance	18100 students	57707 students	26130 students	30886 students
Average Waiting Time for students without the ones in the queue	235876.8 min	301594 min	345490.62 min	106684.86 min
Average adjusted Waiting Time for students	235411 min	300808 min	291514.9 min	106797.4 min 8
Average Turnaround Time for students	235877.66 min	301595.47 min	345495 min	106686 min

Total student arrived	133365 students	333189 students	199936 students	500158 students
Total students served	70723 students	133167 students	109527 students	392995 students
Students served percentage	53%	39%	54%	78%
Student Throughput	0.07 students/min	0.13 students/min	0.1 students/min	0.39 students/min
Professor Queue Average	0 profs	0 profs	0	0
Professor Queue Maximum	1 profs	1 profs	3 min	0
Professor Queue Variance	0.08 profs	0.09 profs	0.12 min	0
Average Waiting Time for Professors without the ones in the queue	0.05 min	0.05 min	0.029 min	0
Average adjusted Waiting	0.05 min	0.05 min	0.029 min	0

Time for Professors				
Average Turnaround Time for Professors	11.05 min	8.05 min	2.03 min	8.5 min
Total Professor arrived	133476 profs	166619prof s	500154 profs	142871 profs
Total Professors served	133475 profs	166618prof s	500151 profs	142869 profs
Professors served percentage	100%	100%	100%	100%
Professor Throughput	0.13 profs/min	0.16 profs/min	0.5 profs/min	0.14 profs/min
Server 0 student Utilization	26%	25%	39%	36%
Server 0 professor Utilization	73%	74%	60%	63%
Server 1 student Utilization	26%	40%	60%	41%
Server 1 professor Utilization	73%	59%	39%	58%

ورودی ۱: زمان ورود اساتید بین ۵ تا ۱۰، زمان ورود دانشجو بین ۵ تا ۱۰، سرویس اساتید بین ۱۰ تا ۱۲، سرویس دانشجو بین ۵ تا ۱۰ و زمان شبیه سازی ۱۰۰۰/۰۰۰ واحد.

```
-----Entered-----
total students entered: 133365          total profs entered: 133476
-----Served-----
Served Profs: 133475          Served Profs percentage: 99.99925
Served Students: 70723          Served Students percentage: 53.029655
Served total: 204198          Served total percentage: 76.52422
-----Throughput-----
Profs Throughput: 0.133475
Students Throughput: 0.070723
total Throughput: 0.204198
-----AvgQueueLength-----
Average Student Queue length: 31395.0
max Student Queue length: 62643.0
variance Student Queue length: 18100.1
Average Prof Queue length: 0.0
max Prof Queue length: 1.0
variance Prof Queue length: 0.083632424
-----ServerUtilization-----
server1 Prof utilization percentage: 0.734452          server1 Student utilization percentage: 0.265543
server2 Prof utilization percentage: 0.734309          server2 Student utilization percentage: 0.265683
server1 utilization percentage: 0.999995          server2 utilization percentage: 0.999992
-----WaitingTime-----
Prof WaitingTime: 0.052773926          Student WatingTime: 235876.8
Prof WaitingTimeAdjusted: 0.05277353          Student WatingTimeAdjusted: 235411.84
-----TurnAroundTime-----
Profs TurnAroundTime: 11.056775
Students TurnAroundTime: 235877.66
```

ورودی ۲: زمان ورود اساتید بین ۵ تا ۷، زمان ورود دانشجو بین ۲ تا ۴، سرویس اساتید بین ۴ تا ۱۲، سرویس دانشجو بین ۴ تا ۶، زمان شبیه سازی ۱۰۰۰/۰۰۰ واحد

```
-----Entered-----
total students entered: 333189          total profs entered: 166619
-----Served-----
Served Profs: 166618          Served Profs percentage: 99.9994
Served Students: 133167          Served Students percentage: 39.967407
Served total: 299785          Served total percentage: 59.980034
-----Throughput-----
Profs Throughput: 0.166618
Students Throughput: 0.133167
total Throughput: 0.299785
-----AvgQueueLength-----
Average Student Queue length: 100142.0
max Student Queue length: 200021.0
variance Student Queue length: 57707.562
Average Prof Queue length: 0.0
max Prof Queue length: 1.0
variance Prof Queue length: 0.09319984
-----ServerUtilization-----
server1 Prof utilization percentage: 0.742624          server1 Student utilization percentage: 0.257374
server2 Prof utilization percentage: 0.590664          server2 Student utilization percentage: 0.409331
server1 utilization percentage: 0.999998          server2 utilization percentage: 0.999995
-----WaitingTime-----
Prof WaitingTime: 0.052593358          Student WatingTime: 301594.97
Prof WaitingTimeAdjusted: 0.05259304          Student WatingTimeAdjusted: 300808.06
-----TurnAroundTime-----
Profs TurnAroundTime: 8.054652
Students TurnAroundTime: 301595.47
```

ورودی ۳: زمان ورود اساتید بین ۱ تا ۳، زمان ورود دانشجو بین ۴ تا ۶، سرویس اساتید بین ۱ تا ۳، سرویس دانشجو بین ۷ تا ۹، زمان شبیه سازی ۱۰۰۰/۰۰۰ واحد

```
-----Entered-----
total students entered: 199936          total profs entered: 500154
-----Served-----
Served Profs: 500151          Served Profs percentage: 99.9994
Served Students: 109527      Served Students percentage: 54.781033
Served total: 609678          Served total percentage: 87.085655
-----Throughput-----
Profs Throughput: 0.500151
Students Throughput: 0.109527
total Throughput: 0.609678
-----AvgQueueLength-----
Average Student Queue length: 45115.0
max Student Queue length: 90409.0
variance Student Queue length: 26130.293
Average Prof Queue length: 0.0
max Prof Queue length: 3.0
variance Prof Queue length: 0.12191496
-----ServerUtilization-----
server1 Prof utilization percentage: 0.602146          server1 Student utilization percentage: 0.397853
server2 Prof utilization percentage: 0.398889          server2 Student utilization percentage: 0.601107
server1 utilization percentage: 0.999999          server2 utilization percentage: 0.999985
-----WaitingTime-----
Prof WaitingTime: 0.029773008          Student WaitingTime: 345490.62
Prof WaitingTimeAdjusted: 0.02977083          Student WaitingTimeAdjusted: 291514.9
-----TurnAroundTime-----
Profs TurnAroundTime: 2.0312345
Students TurnAroundTime: 345499.75
```

ورودی ۴: زمان ورود اساتید بین ۱ تا ۳، زمان ورود دانشجو بین ۲ تا ۴، سرویس اساتید بین ۷ تا ۱۰، سرویس دانشجو بین ۱ تا ۳، زمان شبیه سازی ۱۰۰۰/۰۰۰ واحد

```
-----Entered-----
total students entered: 500158          total profs entered: 142871
-----Served-----
Served Profs: 142869          Served Profs percentage: 99.9986
Served Students: 392995      Served Students percentage: 78.574165
Served total: 535864          Served total percentage: 83.33434
-----Throughput-----
Profs Throughput: 0.142869
Students Throughput: 0.392995
total Throughput: 0.535864
-----AvgQueueLength-----
Average Student Queue length: 53414.0
max Student Queue length: 107165.0
variance Student Queue length: 30886.443
Average Prof Queue length: 0.0
max Prof Queue length: 0.0
variance Prof Queue length: 0.0
-----ServerUtilization-----
server1 Prof utilization percentage: 0.633583          server1 Student utilization percentage: 0.366412
server2 Prof utilization percentage: 0.58084          server2 Student utilization percentage: 0.419153
server1 utilization percentage: 0.999994          server2 utilization percentage: 0.999988
-----WaitingTime-----
Prof WaitingTime: 0.0          Student WaitingTime: 106684.86
Prof WaitingTimeAdjusted: 0.0          Student WaitingTimeAdjusted: 106797.484
-----TurnAroundTime-----
Profs TurnAroundTime: 8.500228
Students TurnAroundTime: 106686.86
```